
Causal Inference Project: Structural Causal Model for Adversarial Meta-Learning

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1 Motivation

Adversarial manipulation of human decision-making is a growing concern in settings where automated agents interact repeatedly with human users. In the multi-round trust game, an investor repeatedly decides how much to invest, while an adversarial trustee or agent decides how much to return; this creates a sequential setting in which trust, exploitation, and learning evolve over time. Prior work by Dezfouli et al. shows that reinforcement learning agents can discover and exploit systematic vulnerabilities in human sequential decision-making in such repeated interaction tasks [1]. Related work on repeated and multi-round trust games also shows that repeated interaction is useful for studying trust, reciprocity, and strategic behavior over time [3, 4]. Building on this literature, our research question is: does communication between adversarial agents causally affect investor return? We define the outcome Y as the investor’s cumulative return and the treatment(Z) as:

$$Z = \begin{cases} 1 & \text{means that adversarial agents can communicate with each other} \\ 0 & \text{means that they cannot.} \end{cases}$$

This question is causally interesting because the validity of standard causal assumptions depends on the structure of the data-generating process. In the language of the course, some of the main structural assumptions are ignorability, SUTVA, and overlap. Under SUTVA, one unit’s potential outcome should not depend on the treatment or behavior of another unit. However, this assumption can fail in networked or group settings where units affect one another. Hudgens and Halloran formalize this problem as causal inference with interference, where one unit’s outcome may be affected by another unit’s treatment assignment [2]. Our setting creates a similar issue: if investors or adversarial agents communicate across pairs, then investor return in one pair may depend on information or behavior from the other pair.

We therefore focus on a simulation-based design that evaluates the same treatment under two different data-generating processes, as illustrated in Figure 3. We then compare the Estimated Treatment Effect (ATE) of agent communication in the two DGPs:

$$\tau_0 = \mathbb{E}[Y(1, 0) - Y(0, 0)] \quad \text{and} \quad \tau_1 = \mathbb{E}[Y(1, 1) - Y(0, 1)],$$

where the first argument denotes agent communication and the second denotes investor communication. This design was chosen because it directly evaluates how a causal estimate behaves when the same treatment is studied under one setting where the key assumption is fulfilled and another where it is violated.

Evaluating a Causal Inference Design Across Two Data-Generating Processes (DGPs)

Treatment, Z = agent communication | Outcome, Y = investor return | Estimand: $E[Y(1) - Y(0)]$

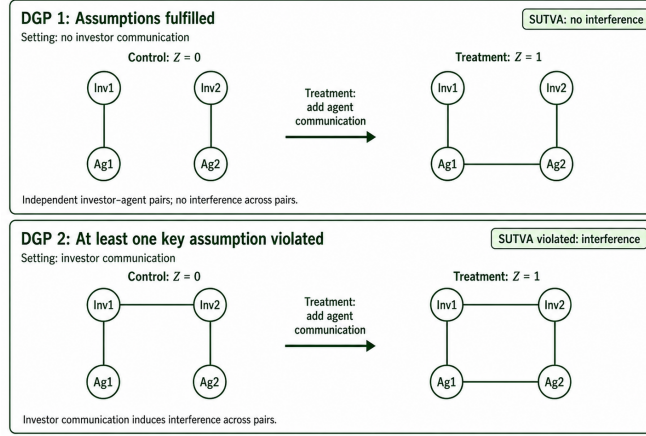


Figure 1: Overview of the MRTT Simulation (DGP).

2 Assumptions

The project requirement is to evaluate the same causal design under at least two data-generating processes, where the relevant assumptions are fulfilled in one setting and at least one key assumption is violated in another. We therefore construct two DGPs that differ only in the presence of investor communication. (Fig 3)

Most assumptions are held fixed by design across the two DGPs.

- Ignorability: satisfied by simulation because Z is assigned by the experimenter rather than selected by agents or investors. There is no self-selection into treatment.
- Overlap: satisfied by design because we explicitly generate observations under both treatment conditions.

Thus, the comparison between $Z = 1$ and $Z = 0$ is not driven by lack of common support. These assumptions are not the source of contrast between the two DGPs.

The key assumption that changes across the two DGPs is SUTVA, specifically the no-interference component. In DGP 1, there are two independent investor-agent pairs,

$$(I_1, A_1), \quad (I_2, A_2),$$

with no investor communication. The return for investor I_1 depends only on the interaction between I_1 and A_1 , and the return for investor I_2 depends only on the interaction between I_2 and A_2 . Because the two pairs are isolated, one pair's outcome is not affected by the other pair's information or behavior. In this setting, SUTVA is fulfilled.

In DGP 2, investors communicate with each other. Now the return for investor I_1 may depend on information transmitted from investor I_2 's interaction with adversary A_2 , and vice versa. This creates cross-pair interference. Therefore, the potential outcome for one investor-agent pair can be affected by the information or behavior of another pair. In this setting, SUTVA is violated.

This gives the required contrast. The design holds ignorability and overlap fixed by simulation, while varying the DGP in a way that preserves SUTVA in one setting and violates SUTVA in the other. Therefore, any difference in the behavior of the estimated treatment effect across the two DGPs can

be interpreted as reflecting the impact of interference generated by investor communication, rather than changes in treatment assignment or support.

3 Design/Estimator

Given our design, the causal effect of interest compares the counterfactual outcomes $Y(1)$ and $Y(0)$, since causal inference requires comparing what would happen under treatment to what would happen under control. In the course notation, our main target is the average treatment effect (ATE).

This follows the standard potential-outcomes framing from class, where causal inference is treated as a missing data problem because both potential outcomes cannot be observed for the same unit at the same time. In our simulation, however, we can approximate these counterfactual comparisons by repeatedly generating outcomes under both treatment conditions. The goal of the design is therefore to create fair comparisons between $Z = 1$ and $Z = 0$ while varying the data-generating process in a controlled way.

The estimator is the difference in mean simulated outcomes. For a fixed data-generating process d , we simulate $R = 50000$ independent replications under $Z = 1$ and $R = 50000$ independent replications under $Z = 0$. Let $Y_{d,1}^{(r)}$ denote investor return in replication r under treatment and $Y_{d,0}^{(r)}$ denote investor return in replication r under control. The estimated effect of agent communication within DGP d is

$$\hat{\tau}_d = \frac{1}{R} \sum_{r=1}^R Y_{d,1}^{(r)} - \frac{1}{R} \sum_{r=1}^R Y_{d,0}^{(r)}.$$

This estimator is used when the researcher can explicitly generate or simulate both treatment and control conditions from a known data-generating process. It is especially useful here because the system is sequential and interactive: investors and agents update their behavior across rounds, and communication can change the structure of dependence between units. Rather than relying on an observational comparison between naturally treated and untreated groups, we intervene directly on the communication structure and compare outcomes across repeated simulated trials.

$$\hat{\tau}_0 = \frac{1}{R} \sum_{r=1}^R Y^{(r)}(1, 0) - \frac{1}{R} \sum_{r=1}^R Y^{(r)}(0, 0),$$

$$\hat{\tau}_1 = \frac{1}{R} \sum_{r=1}^R Y^{(r)}(1, 1) - \frac{1}{R} \sum_{r=1}^R Y^{(r)}(0, 1).$$

Comparing $\hat{\tau}_0$ and $\hat{\tau}_1$ allows us to evaluate whether the estimated effect of agent communication changes when the data-generating process violates SUTVA through investor communication. Thus, the design is not only estimating the effect of Z ; it is also evaluating the stability of that estimate across a setting where the assumptions are fulfilled and a setting where at least one key assumption is violated.

This differs from more traditional observational methods such as regression, matching, weighting, or propensity score adjustment. In those methods, the researcher usually observes one treatment condition per unit and must adjust for differences between treated and control groups. The course notes emphasize that ignorability is an assumption in observational studies: after conditioning on measured pre-treatment covariates X , the potential outcomes must be independent of treatment assignment. This is closer in spirit to a randomized experiment or constructed simulation study than to a standard observational study.

Our main estimator is simply the difference in mean simulated cumulative investor returns, so it does not rely on the parametric assumptions required by a linear regression estimator. This is important because we observe both potential outcomes and the underlying system is nonlinear and sequential: agent behavior, investor state, and communication can evolve across rounds. A regression model could be added later for descriptive or precision purposes, but it is not the primary estimator.

In the simulation, implementation proceeds in four steps. Step 1-3 are clear from the Figure 3, fourth step, we record investor return for each replication and compute the difference in mean investor return between treatment and control. The final comparison is therefore between $\hat{\tau}_0$ and $\hat{\tau}_1$, which shows how the same causal design behaves when the no-interference assumption is satisfied versus when it is violated.

4 Target Estimand

The target estimand is the average treatment effect of adversarial-agent communication on investor return. We may also compare the two quantities (τ_1, τ_0) through

$$\Delta = \tau_1 - \tau_0,$$

which measures whether the causal effect of agent communication changes when investor communication creates interference.

Colloquially, τ_0 asks: when investors do not communicate with each other, how much does allowing adversarial agents to communicate change investor return on average? Similarly, τ_1 asks: when investors do communicate with each other, how much does allowing adversarial agents to communicate change investor return on average? We focus on an average treatment effect rather than an ATT or ATC because treatment is assigned by simulation rather than self-selected by units. There is no naturally treated group whose experience is more substantively important than the control group, and both treatment conditions are generated by design. Therefore, the appropriate estimand is the average effect of turning agent communication on versus off within each DGP.

5 Game Set-up and Simulation Design

5.1 Features of the Design Being Evaluated

The simulation forms a 2×2 design over two communication mechanisms, (Figure 3)

$$(e_{ii}, e_{aa}) \in \{0, 1\} \times \{0, 1\}.$$

Within each investor-communication setting, the estimand is the average effect of turning on adversary communication:

$$\tau_{e_{ii}} = \mathbb{E}[Y(e_{ii}, 1) - Y(e_{ii}, 0)].$$

Thus, the treatment contrast is always between adversary communication and no adversary communication, while the DGP changes according to whether investors communicate.

The main feature of the design we are evaluating is sensitivity to the no-interference component of SUTVA. The central question is whether the same difference-in-means estimator behaves differently when the DGP satisfies no-interference versus when it violates no-interference through investor-side communication.

We are also not primarily evaluating parametric assumptions. The estimator is a difference in mean simulated outcomes, not a linear regression or propensity-score model. Therefore, assumptions such as linear functional form, homoskedasticity, normality of errors, or independence of regression errors are not the central identifying assumptions of the design. The technical environment, payoff equations, RNN investor update, trustee state, and trustee reward structures are described in Appendices B–F. The main text focuses on the causal structure of the simulation.

5.2 Data-Generating Processes

Both data-generating processes contain two investor-trustee dyads:

$$(I_1, A_1), \quad (I_2, A_2),$$

where I_k denotes investor k and A_k denotes adversarial trustee k . Each dyad plays a repeated trust-game interaction. The details of how investment, repayment, investor return, and trustee reward are computed are given in Appendix B.

DGP 1: No investor communication. ($e_{ii} = 0$) The two investor-trustee dyads are isolated with respect to investor-side information. The interaction history of (I_1, A_1) does not affect the investor state or return of (I_2, A_2) , and the interaction history of (I_2, A_2) does not affect the investor state or return of (I_1, A_1) . Within this DGP, we simulate both adversary-communication conditions:

$$e_{aa} = 0 \quad \text{and} \quad e_{aa} = 1.$$

The causal effect of adversary communication in DGP 1 is τ_0 . This is the setting in which the no-interference component of SUTVA is fulfilled with respect to investor-side cross-dyad dependence.

DGP 2: Investor communication. ($e_{ii} = 1$) The state or behavior of investor I_1 may now depend on information from (I_2, A_2) , and the state or behavior of investor I_2 may depend on information from (I_1, A_1) . Within this DGP, we again simulate both adversary-communication conditions.

The causal effect of adversary communication in DGP 2 is τ_1 . This setting violates SUTVA because the outcome for one dyad can be affected by information from the other dyad. Thus, potential outcomes are no longer isolated within each investor-trustee pair.

The comparison between DGP 1 and DGP 2 is the central simulation contrast. Ignorability and overlap are kept fixed by design in both settings. What changes is whether the no-interference assumption is fulfilled or violated. Therefore, differences between $\hat{\tau}_0$ and $\hat{\tau}_1$ evaluate how sensitive the estimated effect of adversary communication is to interference generated by the data-generating process.

5.3 Simulation Parameters

For each of the four simulation cells, we run at least $R = 1000$ independent episodes for each setting:

$$(e_{ii}, e_{aa}) \in \{0, 1\} \times \{0, 1\}.$$

After all replications are complete, we compute the cell means:

$$\bar{Y}_{00}, \quad \bar{Y}_{01}, \quad \bar{Y}_{10}, \quad \bar{Y}_{11},$$

where the first subscript denotes investor communication e_{ii} and the second denotes adversary communication e_{aa} . The estimated treatment effect in DGP 0/1 are

$$\hat{\tau}_0 = \bar{Y}_{01} - \bar{Y}_{00}, \quad \hat{\tau}_1 = \bar{Y}_{11} - \bar{Y}_{10}$$

We also compute

$$\hat{\Delta} = \hat{\tau}_1 - \hat{\tau}_0,$$

which summarizes how the estimated effect of adversary communication changes when investor communication introduces interference.

6 Simulation Results

Table 1 reports the mean investor return in each of the four simulation cells. In DGP 1, where investors do not communicate, the mean investor return increases from -10.047 under no adversary communication to -1.140 under adversary communication. This gives an estimated treatment effect of

$$\hat{\tau}_0 = 8.907.$$

In DGP 2, where investors communicate, the mean investor return increases from 8.177 under no adversary communication to 16.273 under adversary communication. This gives an estimated treatment effect of

$$\hat{\tau}_1 = 8.096.$$

Thus, adversary communication is associated with higher investor return in both DGPs, but the estimated effect is slightly smaller when investor communication is present. The estimated difference in treatment effects is

$$\hat{\Delta} = \hat{\tau}_1 - \hat{\tau}_0 = -0.811.$$

Table 1: Mean investor return by communication condition.

Condition	e_{ii}	e_{aa}	N	Mean investor return	SD
DGP 1 + Control	0	0	1000	-10.047	28.694
DGP 1 + Treatment	0	1	1000	-1.140	31.404
DGP 2 + Control	1	0	1000	8.177	38.422
DGP 2 + Treatment	1	1	1000	16.273	39.194

Because the project requires evaluating bias, root mean squared error, and confidence interval coverage across simulation iterations, we use a bootstrap-based Monte Carlo procedure. The full 1000-episode sample in each cell is treated as the approximate simulation truth. We then generate bootstrap iterations by resampling episodes within each cell with replacement. In each bootstrap iteration b , we recompute the treatment effect estimate

$$\hat{\tau}_g^{(b)} = \bar{Y}_{g1}^{(b)} - \bar{Y}_{g0}^{(b)},$$

where $g \in \{0, 1\}$ indexes the DGP. We then compare each bootstrap estimate to the corresponding approximate estimand, τ_g .

For each DGP, bias and RMSE are computed as

$$\text{Bias}_g = \frac{1}{B} \sum_{b=1}^B \left(\hat{\tau}_g^{(b)} - \tau_g \right), \quad \text{RMSE}_g = \sqrt{\frac{1}{B} \sum_{b=1}^B \left(\hat{\tau}_g^{(b)} - \tau_g \right)^2}$$

For confidence interval coverage, each bootstrap iteration constructs a normal-approximation 95% confidence interval:

$$\hat{\tau}_g^{(b)} \pm 1.96 \cdot \widehat{\text{SE}}\left(\hat{\tau}_g^{(b)}\right),$$

where the standard error for the difference in means is

$$\widehat{\text{SE}}\left(\hat{\tau}_g^{(b)}\right) = \sqrt{\frac{s_{g1}^2}{n_{g1}} + \frac{s_{g0}^2}{n_{g0}}}.$$

Coverage is the proportion of bootstrap intervals that contain the approximate simulation estimand τ_g .

Table 2 summarizes the simulation performance. In DGP 1, where the no-interference component of SUTVA is fulfilled with respect to investor-side communication, the estimator has very small bias (-0.084), RMSE of 1.350, and 95.1% confidence interval coverage. In DGP 2, where investor communication creates cross-dyad interference and violates no-interference, the estimator also has small bias (-0.039), but RMSE increases to 1.756. Coverage remains close to nominal at 95.4%.

Table 2: Bootstrap-based simulation performance of the difference-in-means estimator.

DGP	SUTVA status	Estimand	Bias	RMSE	95% CI Coverage
DGP 1	Fulfilled	8.907	-0.084	1.350	95.1%
DGP 2	Violated	8.096	-0.039	1.756	95.4%

These results suggest that the difference-in-means estimator remains approximately centered around the target estimand in both DGPs. The bias is close to zero in both the assumption-fulfilled and assumption-violating settings. However, the RMSE is larger in DGP 2 than in DGP 1, indicating that the estimator is more variable when investor communication introduces cross-dyad dependence. This is consistent with the causal structure of the simulation: ignorability and overlap are fixed by design, while the main change across DGPs is the presence of interference. Thus, the simulation does not show substantial bias from the SUTVA violation in this implementation, but it does show increased uncertainty under the interference-generating DGP.

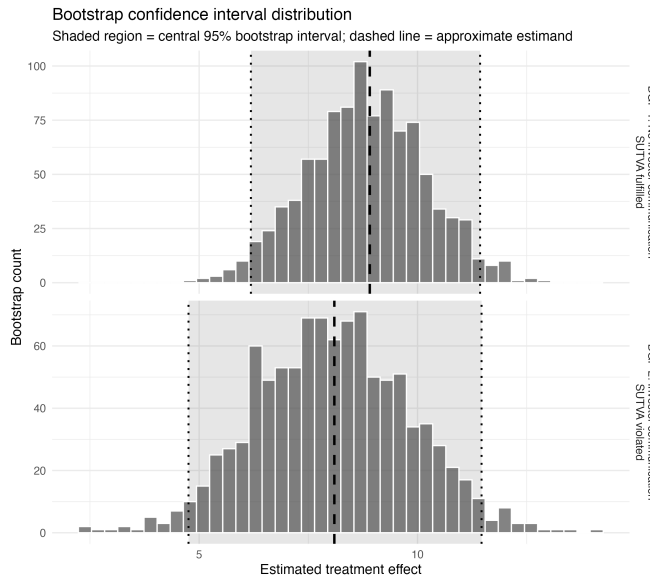


Figure 2: Confidence Interval.

The estimated treatment effects can also be summarized using approximate 95% confidence intervals from the full simulation sample:

$$\hat{\tau}_0 = 8.907 \quad [6.270, 11.543],$$

and

$$\hat{\tau}_1 = 8.096 \quad [4.694, 11.497].$$

The estimated difference in effects is

$$\hat{\Delta} = -0.811 \quad [-5.115, 3.493].$$

Because this interval includes zero, the simulation does not provide strong evidence that investor communication substantially changes the average effect of adversary communication. The clearer difference is in estimator variability: the interference setting has a larger RMSE and larger standard error, suggesting that SUTVA violations may affect the stability of the estimator even when the average bias remains small.

7 Discussion

The simulation results suggest that the difference-in-means estimator performs reasonably well in both data-generating processes, but its stability changes when interference is introduced. In DGP 1, where investors do not communicate and the no-interference component of SUTVA is fulfilled, the estimator is nearly unbiased, has lower RMSE, and achieves close-to-nominal confidence interval coverage. In DGP 2, where investor communication creates cross-dyad dependence and violates no-interference, the estimator remains approximately centered around the target estimand, but the RMSE increases. This suggests that the SUTVA violation in this implementation does not create large average bias, but it does increase estimator variability.

This finding is important because it shows that a violation of a causal assumption does not always appear as dramatic bias in the point estimate. In this simulation, ignorability and overlap were fixed by design, so the main difference between the two DGPs was the presence of investor-side interference. The estimated treatment effect of adversary communication remained positive in both DGPs, but the wider variability in the interference setting suggests that standard difference-in-means comparisons may become less stable when units are connected through communication. In other words, the estimator may still be centered correctly on average, but individual simulation estimates become noisier when outcomes are no longer independent across dyads.

For future methodological choices, this suggests that the simple difference-in-means estimator is appropriate as a baseline when treatment is assigned by design and overlap is guaranteed. However, when interference is substantively plausible, we cannot rely only on a standard estimator that assumes independent units. Instead, explicitly model the interference structure, either by defining treatment at the cluster or network level or by using designs that account for spillovers between units. In this project, comparing DGP 1 and DGP 2 was useful because it made the role of SUTVA visible: the same estimator looked more stable when no-interference held and less stable when communication introduced dependence.

References

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A Description of the Game

The simulation is based on a repeated trust game. In each episode, there are two separate investor-trustee pairs. Each investor starts with a fixed budget and repeatedly decides how much money to invest with their trustee. The invested amount is multiplied by the game, and then the trustee decides how much of the multiplied amount to return to the investor.

For example, if an investor puts in 10 units and the multiplier is 3, the trustee receives 30 units. The trustee can then return any fraction of that amount back to the investor. If the trustee returns a large fraction, the investor benefits and may learn to trust the trustee more in future rounds. If the trustee returns very little, the trustee keeps more for itself, but the investor may reduce future investment.

This repeated interaction creates a setting where trust can grow or decline over time. Investors update their behavior based on what happened in previous rounds, while trustees try to choose repayment strategies that affect investor behavior and final returns.

The main question in this project is whether communication between adversarial trustees changes the investor’s final return. To study this, the simulation has two communication switches. The first switch controls whether the two investors can communicate with each other. The second switch controls whether the two adversarial trustees can communicate with each other.

When investor communication is turned off, each investor-trustee pair behaves like an isolated game. What happens to one investor does not affect the other investor.

When investor communication is turned on, the investors can indirectly share information from their own interactions. This means one investor’s behavior may now be affected by what happened in the other investor-trustee pair.

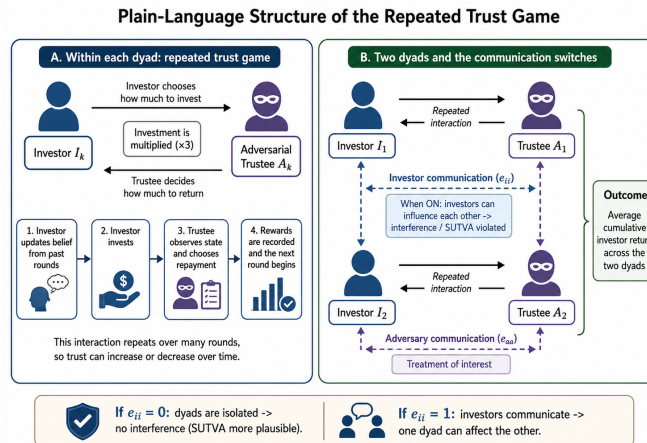


Figure 3: Game Description.

B Environment Notation

There are two investor-trustee dyads indexed by

$$k \in \{1, 2\}.$$

Each dyad contains investor I_k and adversarial trustee A_k . Each episode consists of T repeated rounds. At round t , investor I_k chooses an investment amount $\text{invest}_{k,t}$. This investment is multiplied by a fixed multiplier m , and trustee A_k chooses a repayment proportion $\rho_{k,t} \in [0, 1]$. Repayment is

$$r_{k,t} = \rho_{k,t} \text{invest}_{k,t} m.$$

Investor and trustee rewards are

$$R_{k,t}^I = r_{k,t} - \text{invest}_{k,t},$$

and

$$R_{k,t}^A = \text{invest}_{k,t} m - r_{k,t}.$$

The investor return for dyad k is

$$Y_k = \sum_{t=1}^T R_{k,t}^I,$$

and the episode-level outcome is

$$Y = \frac{1}{2}(Y_1 + Y_2).$$

Each dyad has its own 20-unit wallet. There is no shared budget across dyads and no trust-ranked allocation across agents.

C RNN Investor Model

Each investor is represented by a frozen recurrent neural network. For investor I_k , let

$$h_{k,t} \in \mathbb{R}^5$$

denote the GRU hidden state at round t . Let

$$\hat{a}_{k,t-1}$$

denote the previous investment action encoded as a one-hot vector, and let

$$\rho_{k,t-1}$$

denote the previous repayment proportion. The trained GRU step is written as

$$f(h, \hat{a}, \rho) \rightarrow (h', \pi),$$

where h' is the updated hidden state and π is a five-class policy over investment buckets. The same trained RNN checkpoint is used for both investors, but each investor has its own independent hidden state.

At the start of each round, trust decay is applied:

$$h_{k,t} \leftarrow \gamma h_{k,t}.$$

If investor communication is active, $e_{ii} = 1$, the investors perform a cross-investor update. The hidden states are first snapshotted, and then the update is applied symmetrically:

$$h'_{1,t} = f_h(h_{1,t}, \hat{a}_{2,t-1}, \rho_{2,t-1}),$$

$$h'_{2,t} = f_h(h_{2,t}, \hat{a}_{1,t-1}, \rho_{1,t-1}),$$

where f_h denotes the hidden-state component of the GRU update. Then

$$h_{1,t}, h_{2,t} \leftarrow h'_{1,t}, h'_{2,t}.$$

If $e_{ii} = 0$, this step is skipped.

Each investor then performs its own self update and decision:

$$(h_{k,t}, \pi_{k,t}) = f(h_{k,t}, \hat{a}_{k,t-1}, \rho_{k,t-1}).$$

The desired investment is sampled from $\pi_{k,t}$ or selected by $\arg \max$ under deterministic inference. The actual investment is clipped to the dyad-level wallet:

$$\text{invest}_{k,t} = \text{floor_to_bucket}(\min(\text{desired}_{k,t}, 20)).$$

D Trustee State and Adversary Communication

The trustee state is

$$s_{k,t} = [h_{k,t}, \pi_{k,t}, \text{onehot}(\text{invest}_{k,t}), t/(T-1), e_{ii}, e_{aa}, C_{-k,t}].$$

The cross-window $C_{-k,t}$ is included only when adversary communication is active, $e_{aa} = 1$. It contains the other dyad's previous $d = 4$ investment and repayment pairs:

$$C_{-k,t} = [(\text{invest}_{-k,t-1}, r_{-k,t-1}), \dots, (\text{invest}_{-k,t-4}, r_{-k,t-4})].$$

The window is padded with zeros at the beginning of each episode. If $e_{aa} = 0$, this cross-window is absent rather than zero-filled.

The base state dimension is

$$5 + 5 + 5 + 1 + 2 = 18.$$

When $e_{aa} = 1$, the four-round cross-window adds

$$4 \times 2 = 8$$

additional dimensions, so

$$\dim(s_{k,t}) = \begin{cases} 18, & e_{aa} = 0, \\ 26, & e_{aa} = 1. \end{cases}$$

E Trustee Policies and Reward Structure

The trustees are reinforcement-learning agents that choose repayment proportions. We use two trustee types. The MAX trustee is rewarded by the retained amount,

$$R_{k,t}^{MAX} = R_{k,t}^A,$$

and therefore represents a more exploitative adversary.

The FAIR trustee is constrained toward balance between trustee reward and investor reward. Its step-level reward is zero, and its terminal reward is

$$R_{k,T}^{FAIR} = - \left| \sum_{t=1}^T R_{k,t}^A - \sum_{t=1}^T R_{k,t}^I \right|.$$

This makes the FAIR trustee more reciprocal while still operating in the same sequential environment.

F Bookkeeping and Invariants

After each round, the previous action one-hot and repayment proportion are stored:

$$\begin{aligned} \hat{a}_{k,t} &= \text{onehot}(\text{bucket_index}(\text{invest}_{k,t})), \\ \rho_{k,t} &= \begin{cases} r_{k,t}/(\text{invest}_{k,t}m), & \text{if } \text{invest}_{k,t} > 0, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

These values are used in the next round's self update and, when $e_{ii} = 1$, the cross-investor update.

The construction imposes the dyad-level investment bound

$$0 \leq \text{invest}_{k,t} \leq 20.$$

With multiplier $m = 3$, the reward identity is

$$R_{k,t}^I + R_{k,t}^A = 2\text{invest}_{k,t}.$$

When

$$e_{ii} = 0, \quad e_{aa} = 0,$$

each dyad is equivalent to an independent single-investor, single-trustee repeated trust-game run.